



St. Bonaventure University
QMX-210
Quantitative Applications for Business Students
Spring 2026

Instructor Information

Professor Evelyn Bysiek

- Office Location: De La Roche 301A (Math Suite)
- E-mail address: eabysiek@sbu.edu
- Phone number: (716) 353 – 0666 (cell)
- Office Hours: By Appointment

Vision: The St. Bonaventure University School of Business seeks to be recognized as the premier Franciscan business school, one that excels in business education, applied scholarship, and impactful engagements with the regional, national, and global communities in which we live.

Mission:

Grounded by Franciscan values, the mission of the St. Bonaventure University School of Business is to develop ethical leaders for the common good and the bottom line. We encourage and support our student's development and growth inside and outside the classroom; contribute to the understanding of effective business and educational practice; and work for the betterment of our communities.

Values:

The following Franciscan Values are uniquely linked to our Franciscan mission and are incorporated into our curriculum and services learning initiatives.

Contemplation

To think deeply, to ponder, to meditate, to consider, to study, to plan, to ask “what does it all mean?” is to be a contemplative

Love

To learn ... not for the sake of accumulating knowledge alone, but so that knowledge can be applied to the betterment of others.

Respect

To be a “global village” in which people of all social classes, races, nations and beliefs live, accept, and attempt to understand and support one another.

Joy

To believe that the God who made the world is a good God, that the world and the people in it are good, and that whatever difficulties we face, it's all going to work out for the best. To look at life through a positive lens.

Peace

To uncover the injustices which are the roots of hatred and violence. To prepare ourselves with knowledge, skill and love to promote a better world; to truly be an “instrument” of God's peace.

Compassionate Service

To volunteer, to help, to seek opportunities to be of assistance to others. To give of our time with a sense of humility and a desire to make the world a better place. To see education as a means of being able to serve our fellow human beings.

Course Description

An introduction to quantitative techniques used in business, including real-word application problems, linear equations and inequalities, systems of equations, basic notions from the mathematics of finance, including simple and compound interest, annuities, amortization, graphical methods for solving linear programming optimization tasks, along with an introduction to calculus. 3 credits. Fall and Spring.

BBA Program Objectives:

Course is a required/elective course for all Business majors, and supports the program learning goals and objectives as shown below:

Program Learning Goals	Objectives	Course Activity	Level of Coverage (Mi, Mo, Ex)
Goal 1 – Knowledge of Business Disciplines:	1. Students will demonstrate knowledge of financial formulas involving simple and compound interest. 2. Students will understand how to create linear models of business applications, including depreciation, cost analysis, and markup policy in business. 3. Students will understand how to find a company's revenue, cost, and profit formula and will be able to use these formulas to perform break-even analysis and profit-loss analysis.	Students will model financial problems by applying formulas and quantitative reasoning techniques. Additionally, students will work through a variety of business applications to solve real-life problems.	Ex
Goal 2 – Social Responsibility:	1. Graduates will apply an ethical framework to decision making. 2. Graduates will relate Franciscan values impact to contemporary business practice.	Students will consider quantitative business application problems from an ethical perspective.	Mo
Goal 3 – Critical Thinking:	1. Graduates will demonstrate critical thinking skills and the ability to apply them to solving business problems.	Through the use of algebraic and calculus modeling, students will actively engage in mathematical approaches to making effective business decisions.	Ex
Goal 4 – Globalization & Diversity:	1. Graduates will be able to identify and articulate the impact of contemporary global and multi-cultural issues in business. 2. Students will be able to demonstrate appropriate response to differences in a global environment.	N/A	Mi
Goal 5 – Effective Communication:	1. Graduates will demonstrate effective written communication skills. 2. Graduates will demonstrate effective oral communication skills.	Students will orally respond to critical thinking questions.	Mo
Goal 6 – Technology:	1. Graduates will demonstrate effective use of technology in the areas of word processing, use of spreadsheets, and presentation software.	N/A	Mi

Course Objectives

Upon completion of the course, the student should be able to:

1. Solve linear and quadratic equations algebraically and graphically (*Calculation and Representation*)
2. Set up and solve linear business functions, including profit, revenue, and cost (*Representation and Calculation*)
3. Use function notation (*Calculation*)
4. Model and solve business application word problems using linear or quadratic equations, including break-even analysis, supply and demand, equilibrium points, and profit-loss analysis. (*Application/Analysis, Assumptions, Communication, Interpretation and Calculation*)
5. Use exponential and logarithmic functions in applications (*Calculation*)
6. Model financial problems including simple and compound interest, annuities, amortization, and calculate earned/paid interest by applying basic formulas and reasoning. (*Application/Analysis, Assumptions, Communication, Interpretation and Calculation*)
7. Solve systems of linear equations graphically and algebraically (*Calculation*)
8. Model a linear programming problem (*Representation and Communication*)
9. Solve a linear programming problem graphically (*Calculation, Application/Analysis, and Representation*)
10. Compute derivatives of basic functions (*Calculation*)
11. Apply derivatives as rates of change, slope of the tangent line, and velocity (*Interpretation, Representation, Calculation, Application/Analysis, Communication*)
12. Model optimization problems and use derivatives to solve them (*Application/Analysis, Assumptions, Communication, Interpretation and Calculation*)

Required text:

College Mathematics for Business, Economics, Life Sciences, and Social Sciences,
13th edition, Barnett, Ziegler, and Byleen

Required technology:

A scientific calculator. (Recommended models include TI-30XS, TI-30XIIS, TI-36X Pro, but any scientific calculator is sufficient.) **Note Regarding Calculators:** During quizzes and exams, you may use only an acceptable calculator, i.e. one that is NOT programmable, NOT a graphing calculator, NOT able to store formulas, NOT on a cell phone, and has NO alpha key. **It is your responsibility to be sure that you have an acceptable calculator for quizzes and exams and that you know how to use it.** If you are not sure whether your calculator is acceptable, bring it to me before the day of the first quiz and I will make the determination. For quizzes and exams, sharing of calculators will not be allowed.

University Policies

Academic Honesty Policy at St. Bonaventure University:

Enrollment at St. Bonaventure University requires adherence to the University's standards of academic integrity. These standards may be intuitively understood and cannot, in any case, be listed exhaustively. The following examples, detailed in full in Appendix B of the Code of Conduct (www.sbu.edu/codeofconduct), represent some basic types of unacceptable behavior: cheating, plagiarism, fabrications, obtaining an unfair advantage, aiding and abetting academic dishonesty, falsification of records and official documents, and unauthorized access to records. Academic dishonesty is a serious matter and will be dealt with accordingly, with University sanctions ranging from grade alteration to the possibility of expulsion. Students should familiarize themselves with these very important provisions of the Academic Honesty Policy, which is outlined in the Code of Conduct for reference only. Acts of academic dishonesty are not processed through the University Judicial process; the process for handling alleged violations is outlined within the policy. Please consult the St. Bonaventure University Student Handbook (<http://www.sbu.edu/docs/default-source/about-sbu/code-of-conduct-2016-2017.pdf?sfvrsn=6>) for a list of unacceptable practices, penalties to be assigned and procedures to be followed in prosecuting cases of alleged academic dishonesty. Further, the St. Bonaventure University Undergraduate Catalog may be consulted for specific details on all academic policies at St. Bonaventure.

Students with Disabilities

St. Bonaventure University strives to ensure the accessibility of its programs, classes, and services for all students. Students with disabilities who need academic accommodations should contact the Accessibility Services and Accommodations office, located on the first floor of Plassmann in room 108, at asa@sbu.edu or at 716-375-2115. Additional information about accommodations can be found here: <https://www.sbu.edu/asa>. Please reach out early in the semester so that we can assist you as soon as possible. *Documentation from the Accessibility Services Office is required before the instructor can make accommodations.*

Statement on Student Mental Health

Everyone is on their own mental health journey, but sometimes you can feel like no one understands what you're going through. This is why the University's Center for Student Wellbeing has partnered with Togetherall — an anonymous, online peer-to-peer community where students can connect with others who have real lived experience. The community is free to our students and available 24/7. You must register with your university email to access the platform at <https://account.v2.togetherall.com/register/student>. If you would prefer a more formal experience, counseling services is available on-campus to students and can help you address a variety of concerns. You can schedule an appointment by calling 716-375-2310, option 1, or submit a Counseling Request Form through your patient portal – <https://sbu.medicatconnect.com>. If you find yourself in a crisis, you can walk in and be seen for crisis intervention during regular business hours. You may also contact campus safety & security at 716-375-2525 for assistance after-hours. Other resources are the Crisis Text Line – text HOME to 741741 or the suicide hotline at 988. For additional information, please go to www.sbu.edu/studentwellbeing.

AI Policy

Use of artificial intelligence (AI) tools (such as ChatGPT, Photomath, or similar programs) is strictly prohibited on all graded assignments, including take-home work. All graded work must reflect your own understanding and the methods taught in this course. If a submitted assignment uses methods or solutions that differ significantly from what was demonstrated in class, I may ask you to explain your reasoning in order to receive credit. Failure to demonstrate your own understanding will result in loss of credit for the assignment.

Description of Instructional Methodology

Much of the course will be lecture-based, with a great deal of opportunity for you to practice problems alongside of my lecturing. I will hand out a note packet every time that we begin a new topic, and you will complete the notes and practice problems alongside of my lecturing. If you follow along during class and complete the notes as we go, you will be much more likely to be successful in the homework for that topic, and thus you will perform better on the quizzes and exams. We will also have some classes dedicated to collaboration and group work, where you can further practice application problems in groups.

Instructor Policies

Attendance

Attendance will be taken at every class, and class attendance will be crucial to your success.

Participation

Students are expected to actively participate in class by asking and answering questions, filling out the note packet as we go, and completing practice problems.

Preparation

Students are expected to have homework completed by the beginning of class, and to come prepared with your book, notebook/folder or binder, a pencil, and your scientific calculator.

Courtesy

Students are expected to arrive on time for each class meeting and to refrain from behavior that distracts other students (answering cell phones, texting, watching videos on your phone/computer, etc.).

Collaboration

There will be occasions during this course when students will be given the opportunity to work collaboratively with one other student or in a larger group. Collaborative work allows students to learn from each other. Individual participation is key to the success of any collaboration.

Evaluation Procedures

Breakdown of Grades:

Examinations: 30% of grade

Two examinations, each comprising 15% of your overall grade, will be administered throughout the semester. Failure to show up will result in an automatic zero for the exam.

Homework: 20% of grade

Homework is the most important part in doing well in this class. How do you study for a math class? You do practice problems, i.e. homework. If you do well on your homework, you will do well on tests and quizzes in this class. At the end of almost every class, a homework assignment will be given. The majority of the time, homework will be graded based on completion. **If you answer every question, show work for every question where you can show work, and turn in a quality assignment, you will receive a “3.”**

3: 100% (Homework is completed well)

2: 67% (One or two problems weren't completed, or it is completed late)

1: 33% (1/3 of the assignment was completed and turned in)

0: 0% (Homework was not turned in/cheating occurred)

Quizzes: 15% of grade

There will be approximately 5 quizzes given throughout the semester. The best 4 quiz grades will be used in calculating your final grade (so your lowest quiz grade will be dropped). There are no make-ups for missed quizzes.

Attendance and Preparation: 15% of grade

If you show up to class prepared, act respectful, **stay off of your cell phone**, and engage in the class lecture, you will have a perfect attendance and preparation grade. If you sleep during class, use your cell phone, act disrespectful, don't come prepared, or skip class altogether, you will lose these easy points towards your final grade. Attendance will be taken at each class. So, to summarize: Come to class, be prepared, be respectful, and we will have a great semester!

Post-pandemic, the “attendance” portion of the grade has shifted a bit. **If you are sick in any way – any symptom whatsoever – do not come to class! Please be symptom-free for at least 48 hours!** You will **not** lose points! Every lecture has been put on Moodle. If you cannot come to class, watch the lecture, and let me know you watched it. If you have any questions, make an appointment for Zoom office hours. You can still receive a perfect attendance and preparation grade if you do the work in your room!

****If you cannot come to class due to sickness or an excused absence (i.e. athletic event) – you need to:**

- 1.) E-mail me *prior* to the missed class to inform me you will not be in class,**
- 2.) Watch the video of the material you missed, and**
- 3.) E-mail me AGAIN informing me you watched the video! At this time, I will update your attendance grade!**

Final Exam:**20% of grade**

The final examination will be comprehensive in nature and so you will be expected to answer questions concerning all of the material discussed during this semester. The date of the final exam is TBA.

Grades:	A: 93 – 100	A - : 90 – 92.9	B+ : 87 – 89.9	B : 83 – 86.9
	B - : 80 – 82.9	C + : 76 – 79.9	C : 73 – 75.9	C - : 70 – 72.9
	D+ : 67 – 69.9	D : 63 – 66.9	D - : 60 – 62.9	F : < 60

Course Outline:**QMX 210- Quantitative Applications for Business Students****Spring 2026******** Schedule is Subject to Change!********Tentative Schedule of Topics:**

1/21	Syllabus, Skills Assessment, Introductions	3/16	3.4 – Present Value of an Annuity
1/23	1.1 – Solving Linear Equations and Inequalities	3/18	3.4 – Present Value of an Annuity
1/26	1.1 – Applications of Linear Equations and Inequalities (Intro to Solving Word Problems)	3/20	Chapter 3 Group Work
1/28	1.1 – Applications of Linear Equations and Inequalities (Break-Even Analysis)	3/23	4.1 – Systems of Linear Equations
1/30	1.2 – Lines and Graphs	3/25	4.1 – Applications of Systems of Linear Equations
2/2	1.2 – Applications of Lines and Graphs	3/27	Review for Examination II
2/4	2.1 - Introduction to Functions/Function Notation/Doman and Range	3/30	Examination II
2/6	2.1 – Applications of Functions	4/1	5.1 – Linear Inequalities in Two Variables and 5.2 – Systems of Linear Inequalities in Two Variables
2/9	2.3 – Solving Quadratic Functions	4/3	5.1 – Linear Inequalities in Two Variables and 5.2 – Systems of Linear Inequalities in Two Variables
2/11	2.3 – Applications of Quadratic Functions	4/6	5.3 – Linear Programming Problems
2/13	Group Work	4/8	5.3 – Linear Programming Problems
2/16	Review for Examination I	4/10	5.3 – Linear Programming Problems
2/18	Examination 1	4/13	5.3 – Linear Programming Problems
2/20	3.1 – Simple Interest	4/15	5.3 – Linear Programming Problems
2/23	2.5 – Exponential Functions	4/17	10.4 – Introduction to Calculus and the Derivative
2/25	2.5 – Exponential Functions	4/20	10.5 – Basic Derivative Rules
2/27	2.6 – Logarithmic Functions	4/22	10.5 – Basic Derivative Rules
3/2	No Class – Winter Break	4/24	10.5 – Basic Derivative Rules
3/4	No Class – Winter Break	4/27	10.5 – Basic Derivative Rules
3/6	No Class – Winter Break	4/29	10.7 – Marginal Analysis
3/9	3.2 – Compound and Continuous Compound Interest Applications	5/1	12.6 – Optimization Problems: Maximizing Revenue
3/11	3.3 – Future Value of an Annuity	5/4	12.6 – Optimization Problems: Maximizing Revenue
3/13	3.3 – Sinking Funds	5/6	Review for Final Exam

